Neil A. Hamlett, D.Sc., MBA

Research-Oriented AI & Cyber-Security Consultant

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Summary.

- *Battle-tested, battle-scarred: Described hard-earned pragmatic experience about mission-relevant AI success in recently-published book chapter.*
- *Broad and deep practical experience spanning multiple industry sectors and technology-delivery segments.*
- *System-of-systems perspective: Excels at navigating complex, ambiguous environments.*

Data-Technology Tools and Methods.

- *Python, R, Scala, SQL, JSON, Neo4j, Protobuf.*
- *Probabilistic-reasoning (Bayesian) methods.*
- *Project planning, estimation, management.*
- *Data science, AI in cloud (AWS, Azure, Databricks).*
- *Decision analysis and business-process modeling.*
- *Statistical/machine learning.*
- *Generative AI and natural-language processing.*
- *Agile-development methods.*
- *Model-based system-engineering (UML, DoDAF)*
- *Microservices software-engineering patterns.*

Professional History.

Director of Data Science, Verituity, 2025 – 2026.

Description. *Financial-Technology startup pursuing competitive differentiation in elimination of errors, fraud, mis-directed payments associated digital disbursements.*

Key Accomplishment. *Developed soup-to-nuts payee entity-resolution micro-service intended to improve operational efficiency and strengthen identify verification of Verituity's precision-payout platform. Used AI-coding tools extensively to improve productivity and code quality, and to reinforce knowledge gaps.*

Cyber-Security Data Engineer, ECS Technologies, 2022 – 2025.

Description. *Digital transformation of core activities for U.S. Cybersecurity and Infrastructure Security Agency (CISA) cyber-risk management of mission-critical IT systems across all federal-civilian agencies.*

Key Accomplishments.

- *Assisted High-Value Asset (HVA) Program Management Office (PMO) with migration of core-mission functions from scattered Excel files into an integrated set of Power BI dashboards. Implemented end-to-end data-engineering and -science workflow to integrate information from a variety of sources and data quality.*
- *Shaped federal-wide agenda for cyber-risk management through clarifying OMB-level decisions related managing the federal-civilian portfolio of major IT systems. Researched federal policies, technology-industry practices, and academic literature on enterprise cyber-risk management. Introduced principles leading to more-complete cyber-risk portfolio management, including clearer insights into cyber-risk trends.*

Founder, AI-Implementation Consultant, Uncertainty Research, LLC 2019 – present.

U.S.A.F. Capital-Planning Decision Support, Uncertainty Research, LLC, 2024 – 2025.

Description. *Entrepreneurial initiative led by SNA Software, LLC to explore and implement capital-planning-portfolio decision-support tools to optimize weapons-systems investments. The Air Force seeks to optimize weapons-systems acquisition decisions based on mission contribution. Demonstrated the mission-modeling and decision-science framework to align financial decision-making to nonfinancial criteria. Implemented minimal viable product in SNA's low-code/no-code client-server platform.*

Senior Data Scientist, International Consulting Associates, 2021 – 2022.

Description. *Analytics and artificial intelligence support to U.S. Food and Drug Administration Office of Clinical Evidence and Analysis in Center for Device and Radiological Health (CDRH).*

K**ey Accomplishment** *Developed application to help analysts supporting FDA's medical-device regulatory mission. Incorporated NLP into signal-detection methods employed in FDA's medical-device surveillance.*

***Professional History* (continued).**

Senior Data Scientist, Vectrus, *2020 – 2021.*

Description. *Analytics and artificial intelligence support to U.S. Department of Defense Program Executive for Chemical, Biological, Radiological, Nuclear Defense (PEO-CBRND).*

Key Accomplishment. *Combined physical- and statistical-modeling methods to expand the range of threats to which a CBRN-threat sensor responds. Researched chemistry-industry literature about specific classes of threat sensors. Combined physics-informed model with statistical-detection logic in MVP of sensor-capability extension.*

Lead Education Data Scientist, IBM-Watson Education **(IBM Research),** *2014 – 2019.*

Description. *Product-development leadership role in entrepreneurial artificial-intelligence-based education-technology cloud offering targeting US K-12 market.*

Key Accomplishment. *Created soup-to-nuts the core for the most-compelling AI feature of an ambitious cloud-based personalized-learning application. Built AI for real-time analysis of each student's learning progress using python realization of a Bayesian network. This functionality operationalized best practices in curriculum, instruction, and assessments represented by a broad swath of the education literature.*

Senior Managing Consultant, IBM Global Business Services, *2010 – 2014.*

Description. *Business-development and project-delivery roles for a diverse set of public-sector, private-sector, and nonprofit engagements. Industries included defense, healthcare, education, electronics manufacturing, and telecommunications.*

Key Accomplishments.

- *Designed enterprise data-architecture component of proposal to major federal healthcare provider. End-to-end solution provided foundation for highly granular operations-analytics and reporting of high-value assets throughout the healthcare-delivery system. Received the highest-possible score.*
- *Directed execution of a text-analytics project to mine insights from unstructured data documenting 30 years of material inspections of U.S. Navy warships. Results produced evidence counter to the prevailing assumption that warship readiness necessarily declines with years in service.*

Senior Managing IT Architect, IBM Global Business Services, *2007 – 2010.*

Description. *System-engineering support to business development.*

Key Accomplishments.

- *Designed portfolio-management decision-support system for major U.S. Defense agency. Combined descriptive analytics, constrained-optimization to provide interactive what-if capital-allocation drills. IBM won a multi-year contract with potential value of $10 million.*

Engineering Duty Officer, U.S. Navy, *1993 – 2007.*

Description. *Performed program-management, contract-administration, and system-engineering functions for four major system-acquisition programs in space, surveillance, communications, and networks domains.*

Key Accomplishment. *Devised and executed turnaround for $120 million satellite-telecommunications program previously on trajectory towards cancellation. Oversaw reengineering of critical broadcast-management subsystem employing a novel, elegant optimization approach. Reduced lifecycle cost by $25 million; accelerated mission-critical features by two years.*

Education.

- *General Assembly, Data-Science Immersion,* *2020.*
- *Master of Business Administration, The McDonough School of Business, Georgetown University, Washington, DC,* *2011.*
- *Doctor of Science in Electrical Engineering, The George Washington University, Washington, DC,* *2007.*
- *Master of Science in Electrical Engineering, Naval Postgraduate School, Monterey, CA,* *1993.*

***Professional Societies*.**

- *Society of Decision Professionals.*
- *Strategic-Management Society.*
- *Academy of Management.*
- *Society for Risk Analysis.*
- *Institute for Operations Research and Management Science (INFORMS).*

Professional-Research Interests.

- *Scientific approaches to entrepreneurial and strategic decision-making based on causal inference.*
- *Alignment of technology delivery with strategic, operational, and organizational imperatives.*
- *Applications of data science, artificial intelligence to reasoning and decision-making under uncertainty.*

Professional-Society Leadership Roles.

- *Chair, Program Council, Society of Decision Professionals (SDP),* 2021 – present.
  Leads a council of academic researchers and professional practitioners in programming of series of monthly webinars covering a variety of decision-related topics.
- Co-Chair, Program Committee, *2024 SDP Annual Conference,* 2023 – 2024.
  Led program development for SDP Annual Conference, Arlington, VA April 16-18, 2024, https://m44397societyofdecisionprofessionals.mywebsites360.com/.

Publications, Conference Presentations, and Speaking Engagements.

- *"Evidentiary reasoning with the Analytic Hierarchy Process: Use of graph entropy to rank decision factors." In-progress, May 2026. https://github.com/hamlett-neil-ur/ahp-graph-entropy.*
- *Review committees, Knowledge and Innovation (K&I) Theme Track, Strategic Management Society 45th Annual Conference, San Francisco, CA, October 11-14, 2025, and for the Academy of Management Annual Conference, Philadelphia, PA, August 31-July 4, 2026*
- *A theory and methodology for AI-based digital transformation.* **Achieving Digital Transformation through Analytics and AI**, *J. Leibovitz, ed., World Scientific, https://doi.org/10.1142/13939*
- *"Business of AI" session, Invited Moderator, "Decision Intelligence Confronts AI" virtual workshop, Society of Decision Professionals (SDP), November 2023.*
- *"Bridging Data Science and Decision Analysis", session chair and speaker, INFORMS Annual Meeting, October 2021.*
- *"Explicit calculation of terms in the bias-variance decomposition", Symposium on Data Science and Statistics (SDSS), June 2021.*
- *"Bridging Data Science and Decision Analysis", session chair and speaker, Decision Analysis Affinity Group (DAAG), Society of Decision Professionals, April 2021.*
- *"A Natural-language-processing study of the business-strategy literature spanning 1980-2020", INFORMS Annual Meeting, November 2020.*
- *"Uncertainty in decision-making", session chair, Decision Analysis Affinity Group (DAAG), Society of Decision Professionals, October 2020.*
- *"Data-Governance and Culture", "Data-Quality and Integrity" sessions, invited panelist, Industrial Data Summit (North America), TheManufacturer, September 30, 2020.*
- *"A taxonomy of uncertainty in decision-making", Washington Institute for Operations and Research and Management Science (WINFORMS), March 15, 2020.*
- *"Characterizing uncertainty in decision analysis: A practitioner's perspective", with SDP President Patrick Leach, INFORMS Annual Meeting, Seattle, WA, October 2019.*
- *"Classroom learning as assessment experience: Continuous measurement and diagnosis in mathematics curricula", Special conference on classroom assessment, National Council on Measurements in Education (NCME), Boulder, CO, September 18-19, 2019 (paper accepted but subsequently withdrawn).*
- *"A framework for characterizing decision-making's contribution to competitive differentiation", INFORMS Advances in Decision-Analysis, Milan, IT, June 2019.*
- *"Competitive differentiation based on information," April 2019, unpublished working paper, https://drive.google.com/file/d/1Cq7oM6BYrGWxmjf1bdhrdrORZnKBvLu4.*
- *"Potential artificial-intelligence impact on electronics manufacturing", invited speaker, APEX Expo, Institute of Printed Circuits (IPC), San Diego, CA, January 2019.*
- *"IT outsourcing impacts on enterprise architecture,"* **IT Professional**, *IEEE Computer Society, March 2007, https://doi.org/10.1109/MITP.2007.35.*
- *"A performance baseline for the convergence of electromagnetic integral-equation calculations using pulse functions,* **IEEE Transactions on Antennas and Propagation**, *54(5):1523-1537, May 2006, https://doi.org/10.1109/TAP.2006.872598.*